

VILYUYSK, Russian Federation

WMO: 246410

Lat: 63.783N

Lon: 121.617E

Elev: 364

StdP: 14.50

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-55.8	-52.0	-59.7	0.2	-55.7	-56.0	0.2	-52.1	11.1	-16.0	9.7	-15.0	1.8	240	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	86.7	66.4	82.9	64.5	78.9	62.7	68.0	83.1	65.9	79.8	63.9	76.4	5.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	85.0	73.9	60.3	79.1	72.8	58.2	73.4	71.4	32.5	83.0	31.0	80.0	29.4	76.7	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.0	9.0	-55.8	92.5	4.3	3.5	-58.9	95.0	-61.4	97.1	-63.8	99.0	-66.9	101.6		
(5)			-55.8	70.9	4.3	1.9	-58.9	72.3	-61.4	73.4	-63.8	74.4	-66.9	75.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	-31.0	-22.7	-0.6	23.2	43.0	60.9	65.9	58.4	42.5	19.4	-12.8	-30.6		
	DBStd	36.89	13.65	13.29	14.22	10.76	8.96	9.00	7.45	7.01	8.26	12.21	15.25	13.94		
	HDD50	12784	2511	2035	1568	804	257	17	1	13	250	947	1883	2498		
	HDD65	17268	2976	2455	2033	1254	683	180	82	223	674	1412	2333	2963		
	CDD50	1177	0	0	0	0	39	344	493	275	26	0	0	0		
	CDD65	184	0	0	0	0	0	57	108	19	0	0	0	0		
	CDH74	2078	0	0	0	0	10	702	1153	211	2	0	0	0		
	CDH80	664	0	0	0	0	1	222	402	39	0	0	0	0		
	WSAvg	4.2	3.4	3.3	3.7	4.4	5.2	5.0	4.5	4.3	4.5	4.5	3.8	3.3		
	PrecAvg	10.5	0.4	0.3	0.4	0.4	0.9	1.3	1.9	1.6	1.2	0.9	0.7	0.6		
	PrecMax	16.4	1.1	1.3	0.9	1.2	2.5	2.8	5.0	5.5	3.7	1.9	1.5	1.2		
(15) Precipitation	PrecMin	5.6	0.1	0.0	0.0	0.0	0.0	0.4	0.0	0.2	0.2	0.2	0.2	0.1		
	PrecStd	2.2	0.2	0.2	0.2	0.3	0.6	0.7	1.1	1.1	0.7	0.4	0.3	0.3		
	DB	13.1	19.2	40.3	54.3	75.3	90.5	93.2	84.4	71.6	43.7	30.8	12.5			
	MCWB	12.0	17.3	33.1	42.8	56.1	68.1	70.4	66.0	56.8	38.5	28.3	11.5			
	2% DB	2.3	10.6	32.4	47.8	69.5	85.9	88.3	79.6	65.2	39.9	23.3	1.7			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	1.6	9.4	27.2	38.6	52.5	65.2	67.7	63.1	52.6	36.0	21.6	1.0			
	DB	-4.9	2.7	27.2	43.6	64.1	82.0	84.6	75.6	60.1	36.7	16.5	-4.5			
	MCWB	-5.5	1.8	23.2	35.3	49.4	62.9	65.8	61.0	50.3	33.5	15.4	-5.1			
	DB	-11.3	-3.3	22.2	40.0	59.3	77.6	80.8	71.9	55.6	34.0	10.1	-9.8			
	MCWB	-11.7	-4.0	19.2	33.0	46.8	60.6	63.9	59.1	47.8	31.4	9.1	-10.2			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	12.0	17.6	33.4	43.1	56.6	69.0	71.9	67.4	57.9	39.9	28.8	11.8			
	MCDB	13.2	18.8	40.1	53.3	73.6	87.6	90.0	79.8	69.4	42.8	30.7	12.9			
	WB	1.6	9.4	27.6	39.1	52.8	66.6	69.1	64.8	54.1	36.6	21.6	1.0			
	MCDB	2.4	10.6	32.6	47.3	68.1	83.5	84.6	76.5	61.8	39.1	23.0	1.5			
	WB	-5.5	1.8	23.3	35.8	50.0	64.3	67.2	62.5	51.3	33.8	15.4	-5.1			
(31) Mean Daily Temperature Range	MCDB	-4.9	2.5	27.1	42.9	62.9	79.6	81.3	72.4	59.0	36.3	16.4	-4.6			
	WB	-11.7	-3.9	19.2	33.3	47.2	61.9	65.3	60.3	48.5	31.4	9.1	-10.2			
	MCDB	-11.3	-3.3	22.3	39.4	57.9	75.3	78.1	70.0	55.2	34.1	10.1	-9.8			
	MDBR	11.0	14.9	22.4	21.6	17.5	19.5	18.7	18.3	14.8	10.9	11.4	9.9			
	5% DB	18.7	19.0	21.2	21.8	24.5	26.1	24.8	25.3	23.3	11.3	15.9	16.7			
(36) Clear-Sky Solar Irradiance	MCWBR	18.3	18.4	17.6	14.9	13.7	11.9	9.6	12.0	12.8	9.1	15.3	16.4			
	MCDBR	18.7	18.9	20.7	20.4	23.2	23.7	22.1	20.9	20.5	10.4	15.9	17.0			
	MCWBR	18.3	18.3	17.4	14.3	13.3	11.4	9.6	10.8	12.1	9.0	15.4	16.7			
	taub	0.203	0.252	0.290	0.333	0.372	0.408	0.496	0.473	0.328	0.272	0.199	0.188			
	taud	1.907	2.068	2.095	2.113	2.278	2.285	1.997	2.011	2.448	2.314	1.933	1.881			
(42) All-Sky Solar Radiation	Ebn at Noon	117	211	251	265	265	257	224	215	243	207	116	52			
	Edh at Noon	10	21	31	38	36	37	48	43	23	16	9	4			
	RadAvg	72	364	926	1594	1725	1830	1681	1237	766	310	114	27			
(45)	RadStd	5	17	29	53	100	123	94	102	84	34	11	1			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.58	N/A	N/A	N/A	N/A	N/A	-520	-576	+131	N/A
(47)	Regional (2 neighbors)	+1.40	N/A	N/A	N/A	N/A	N/A	-409	-499	+101	N/A

Nomenclature: See separate page